

Chapter 19

Viruses

PowerPoint® Lecture Presentations for

Biology

Eighth Edition

Neil Campbell and Jane Reece

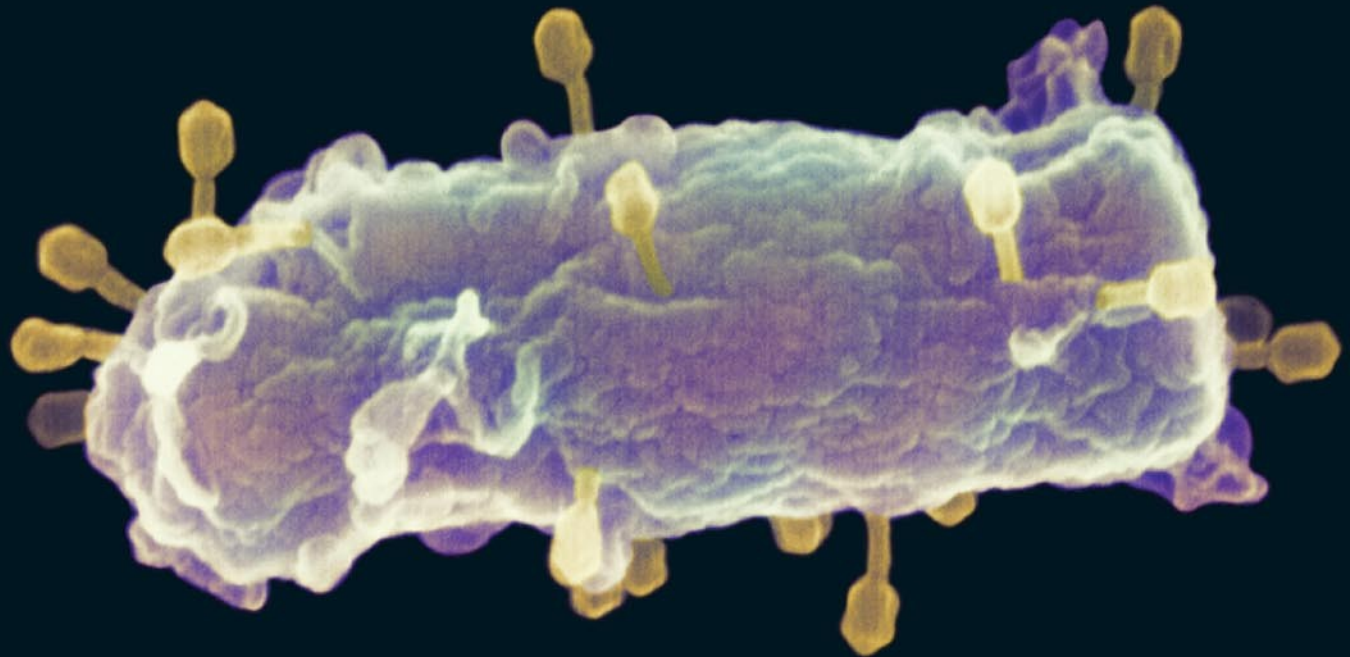
Lectures by Chris Romero, updated by Erin Barley with contributions from Joan Sharp

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Overview: A Borrowed Life

- Viruses called bacteriophages can infect and set in motion a genetic takeover of bacteria, such as *Escherichia coli*
- Viruses lead “a kind of borrowed life” between life-forms and chemicals
- The origins of molecular biology lie in early studies of viruses that infect bacteria

Fig. 19-1



0.5 μm

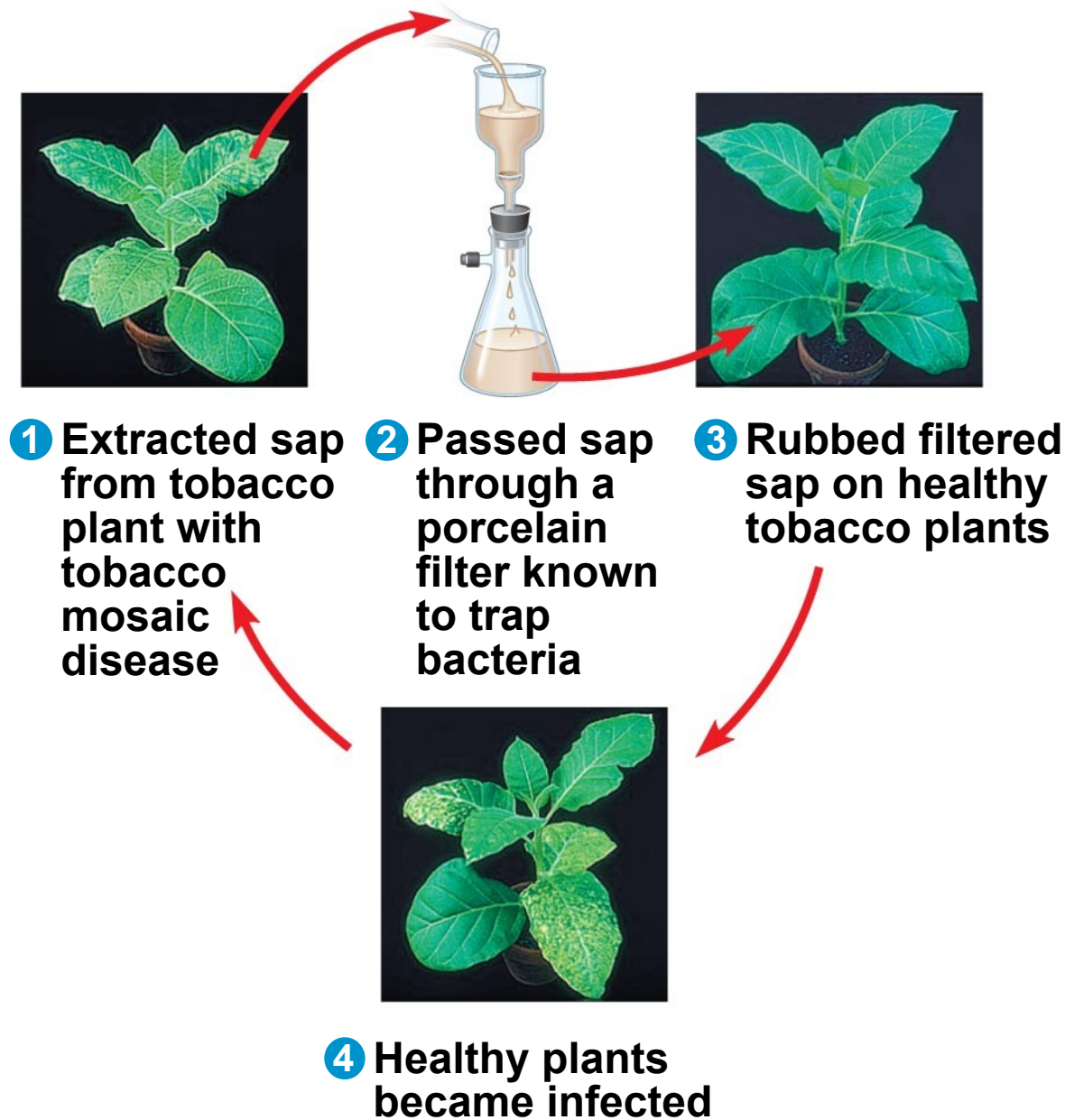
Concept 19.1: A virus consists of a nucleic acid surrounded by a protein coat

- Viruses were detected indirectly long before they were actually seen

The Discovery of Viruses: *Scientific Inquiry*

- Tobacco mosaic disease stunts growth of tobacco plants and gives their leaves a mosaic coloration
- In the late 1800s, researchers hypothesized that a particle smaller than bacteria caused the disease
- In 1935, Wendell Stanley confirmed this hypothesis by crystallizing the infectious particle, now known as tobacco mosaic virus (TMV)

RESULTS



Structure of Viruses

- Viruses are not cells
- Viruses are very small infectious particles consisting of nucleic acid enclosed in a protein coat and, in some cases, a membranous envelope

Viral Genomes

- Viral genomes may consist of either
 - Double- or single-stranded DNA, or
 - Double- or single-stranded RNA
- Depending on its type of nucleic acid, a virus is called a DNA virus or an RNA virus

Capsids and Envelopes

- A **capsid** is the protein shell that encloses the viral genome
- Capsids are built from protein subunits called *capsomeres*
- A capsid can have various structures

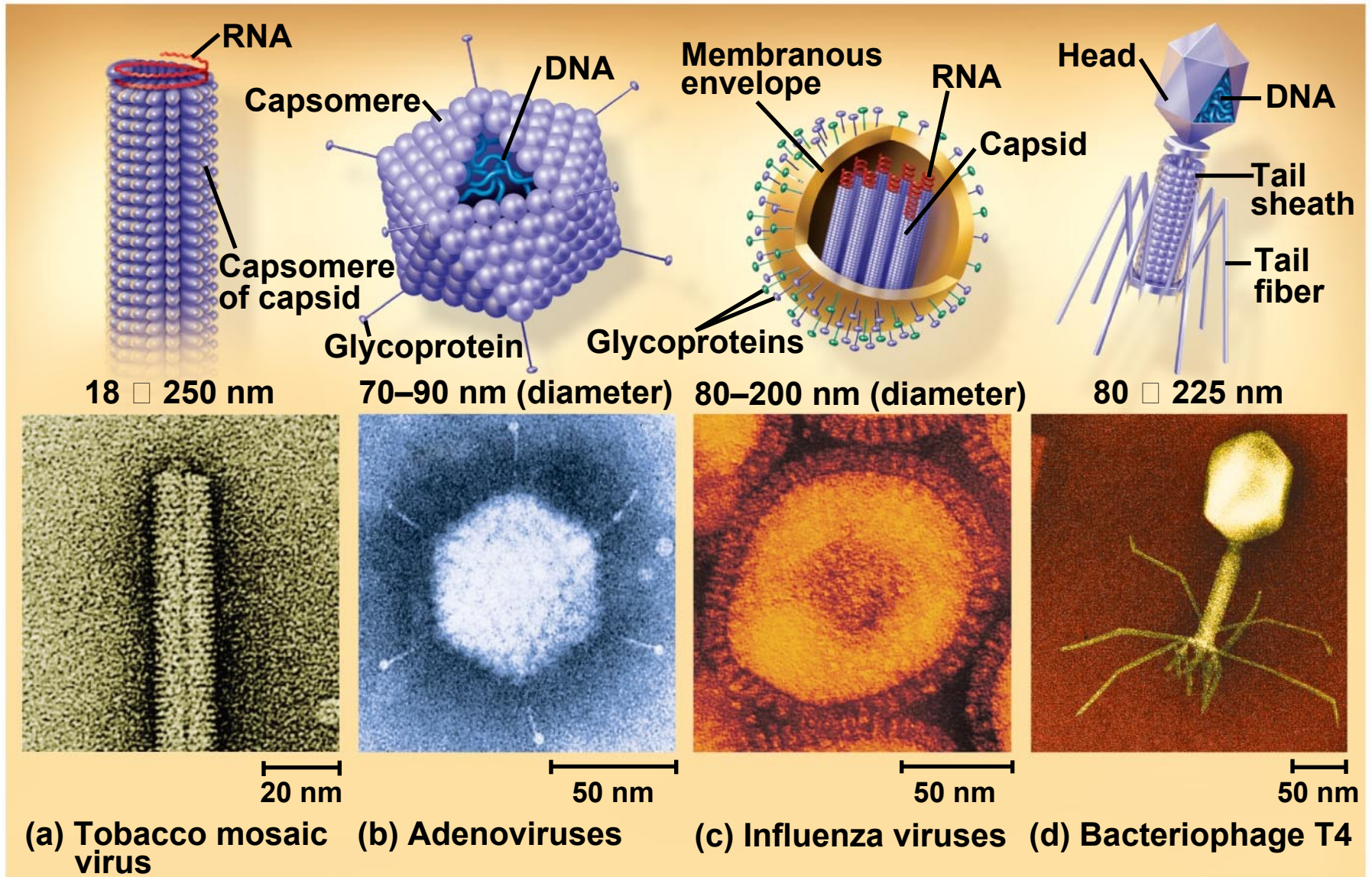
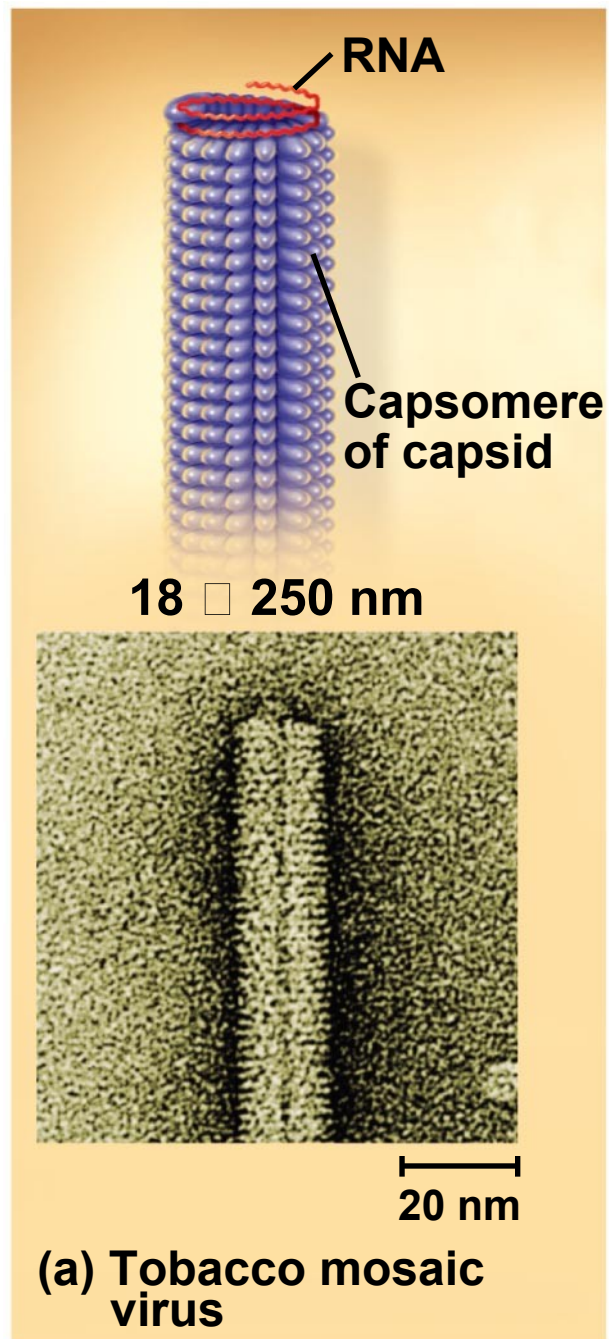
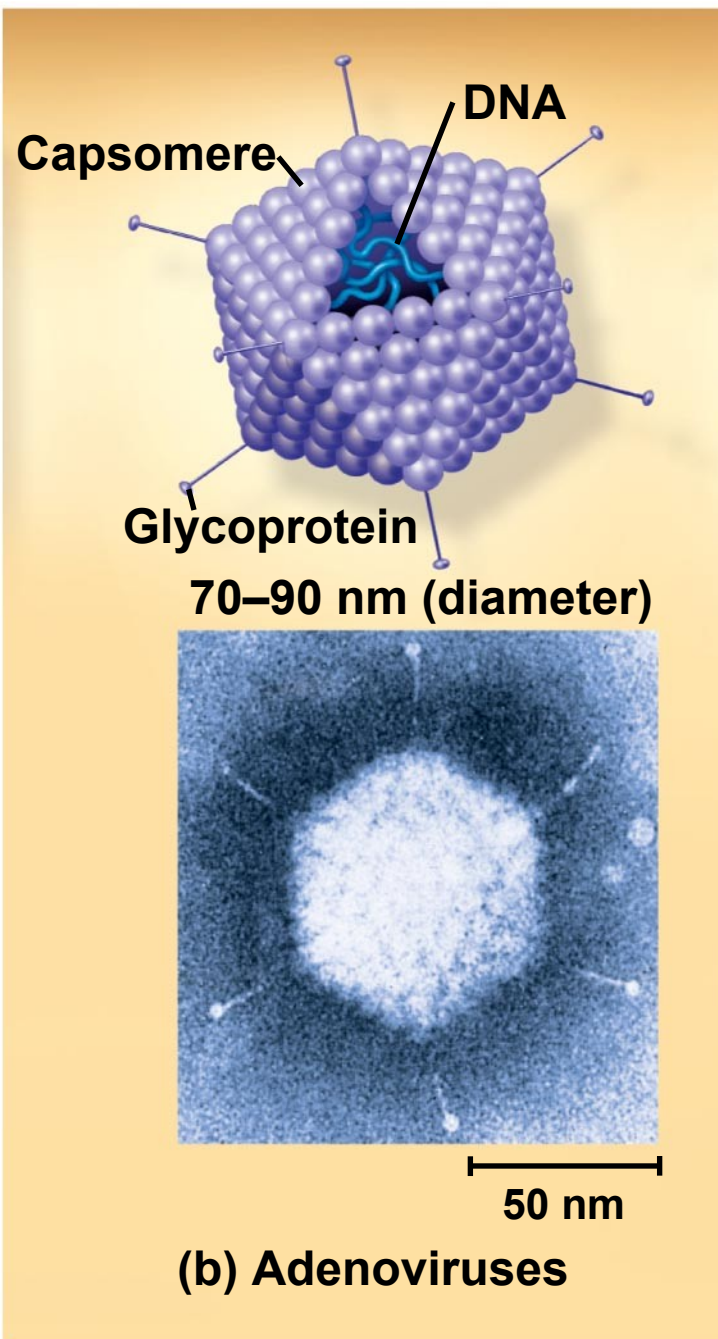
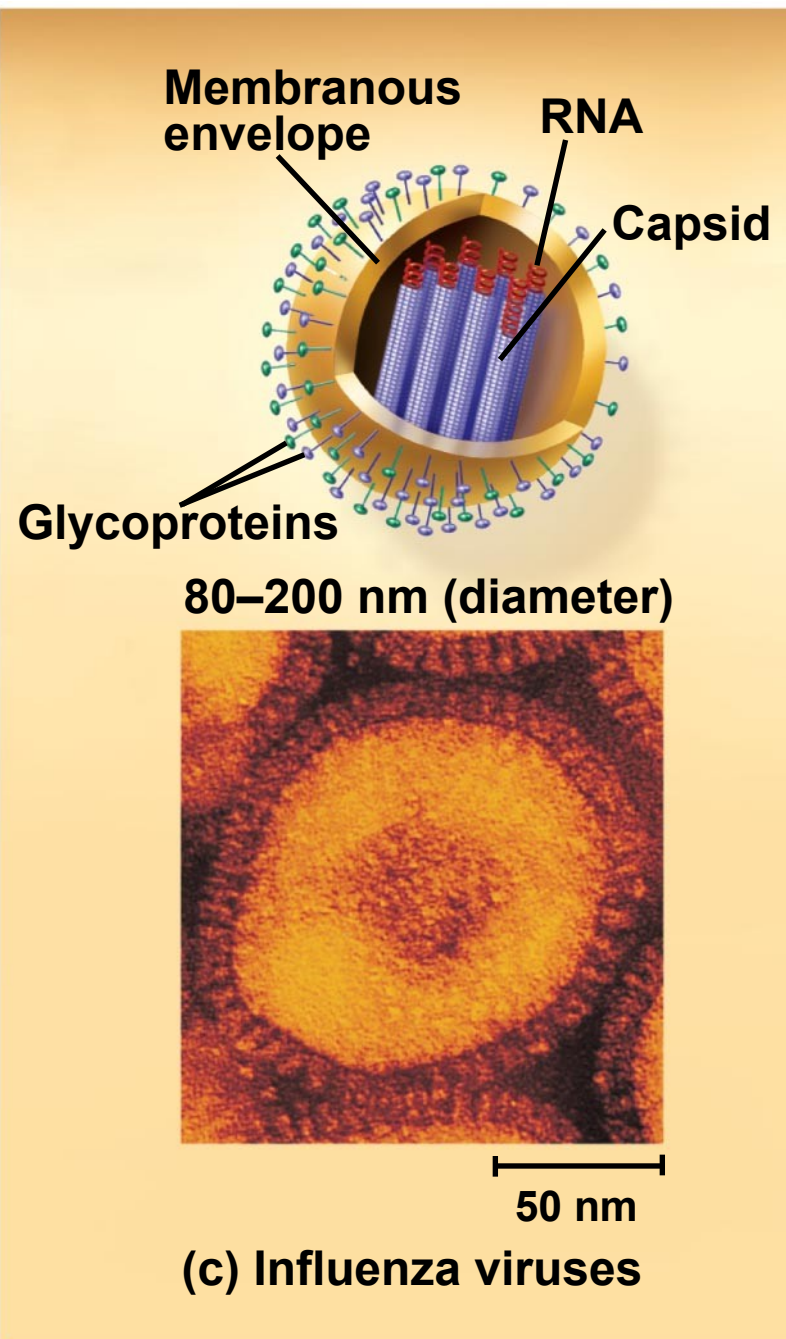
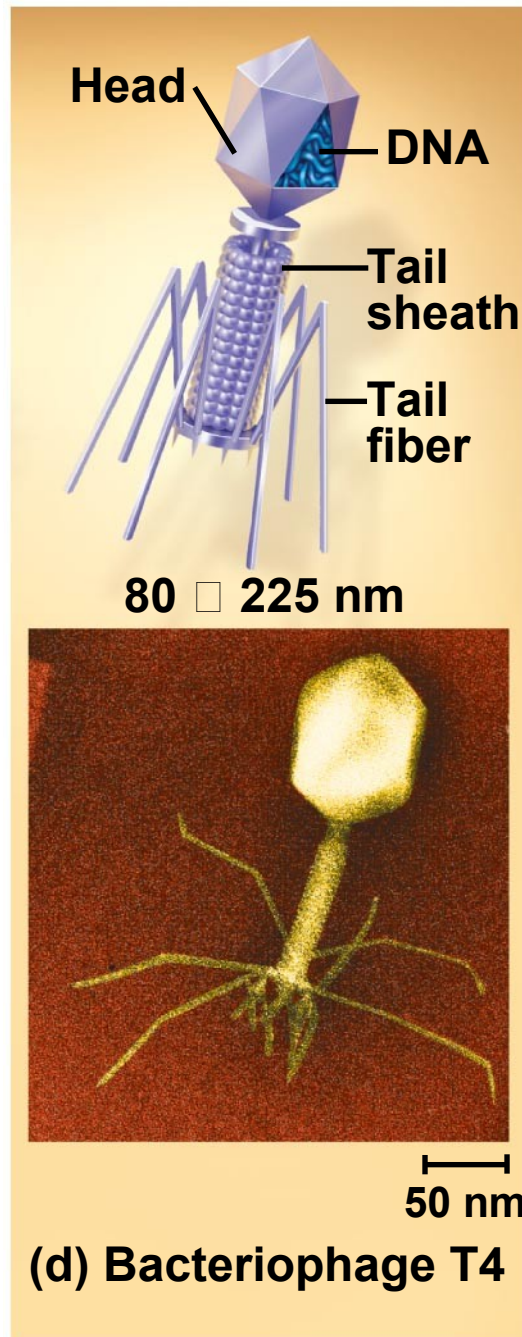


Fig. 19-3a









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- Some viruses have membranous envelopes that help them infect hosts
 - These **viral envelopes** surround the capsids of influenza viruses and many other viruses found in animals
 - Viral envelopes, which are derived from the host cell's membrane, contain a combination of viral and host cell molecules

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- **Bacteriophages**, also called **phages**, are viruses that infect bacteria
 - They have the most complex capsids found among viruses
 - Phages have an elongated capsid head that encloses their DNA
 - A protein tail piece attaches the phage to the host and injects the phage DNA inside

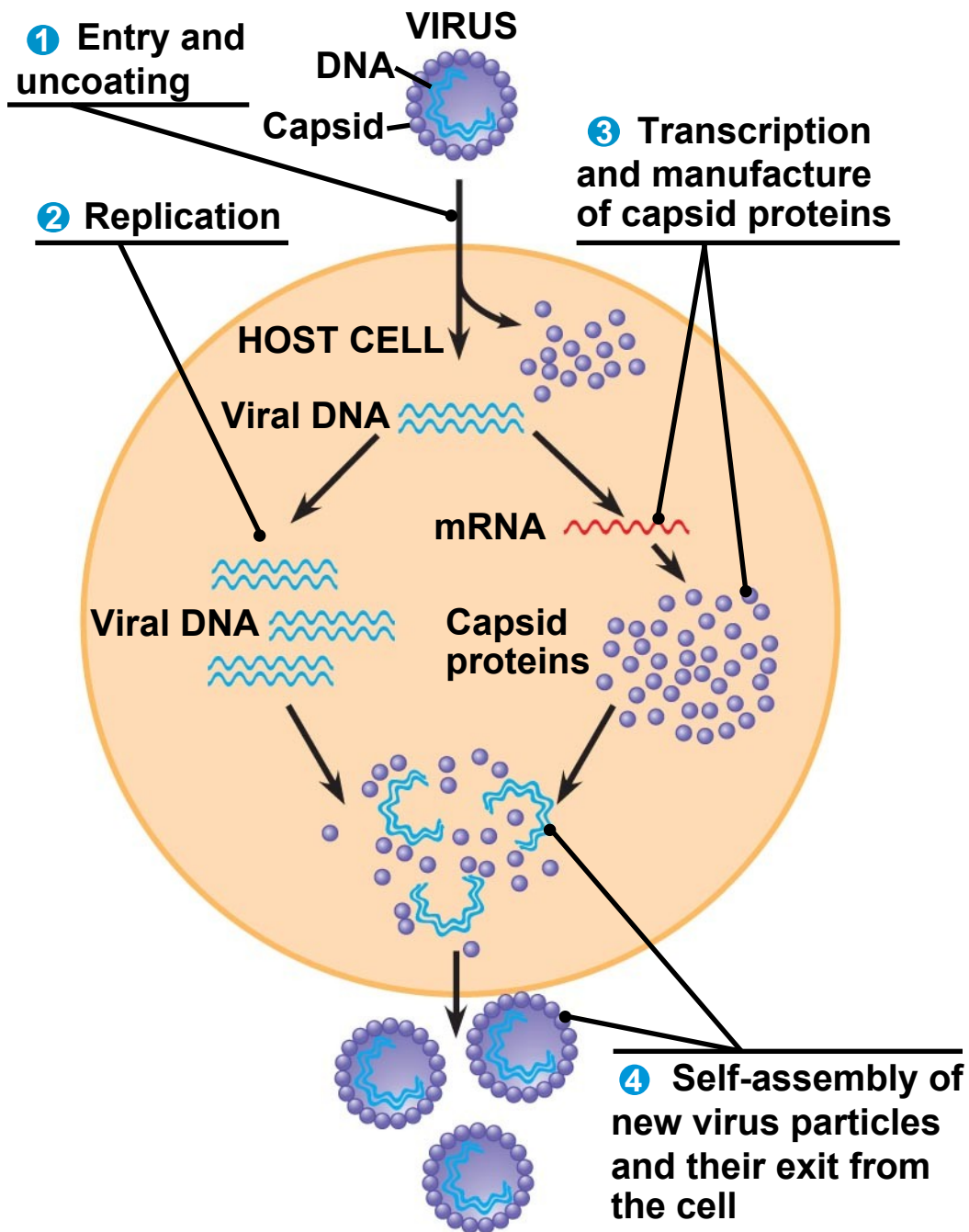
Concept 19.2: Viruses reproduce only in host cells

- Viruses are obligate intracellular parasites, which means they can reproduce only within a host cell
- Each virus has a **host range**, a limited number of host cells that it can infect

General Features of Viral Reproductive Cycles

- Once a viral genome has entered a cell, the cell begins to manufacture viral proteins
- The virus makes use of host enzymes, ribosomes, tRNAs, amino acids, ATP, and other molecules
- Viral nucleic acid molecules and capsomeres spontaneously self-assemble into new viruses

Fig. 19-4



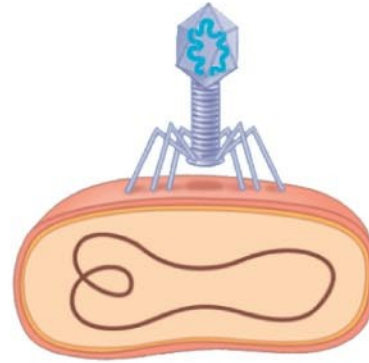
Reproductive Cycles of Phages

- Phages are the best understood of all viruses
- Phages have two reproductive mechanisms: the lytic cycle and the lysogenic cycle

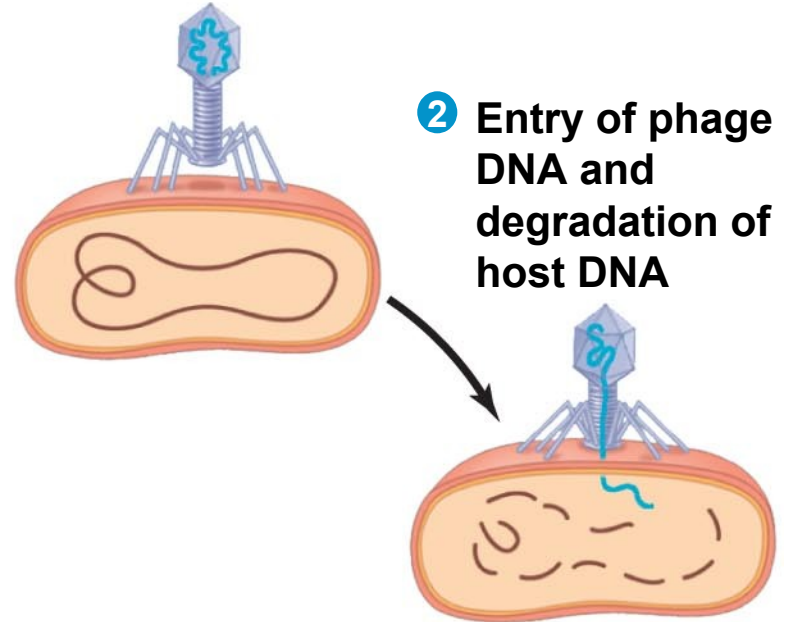
The Lytic Cycle

- The **lytic cycle** is a phage reproductive cycle that culminates in the death of the host cell
- The lytic cycle produces new phages and digests the host's cell wall, releasing the progeny viruses
- A phage that reproduces only by the lytic cycle is called a **virulent phage**
- Bacteria have defenses against phages, including **restriction enzymes** that recognize and cut up certain phage DNA

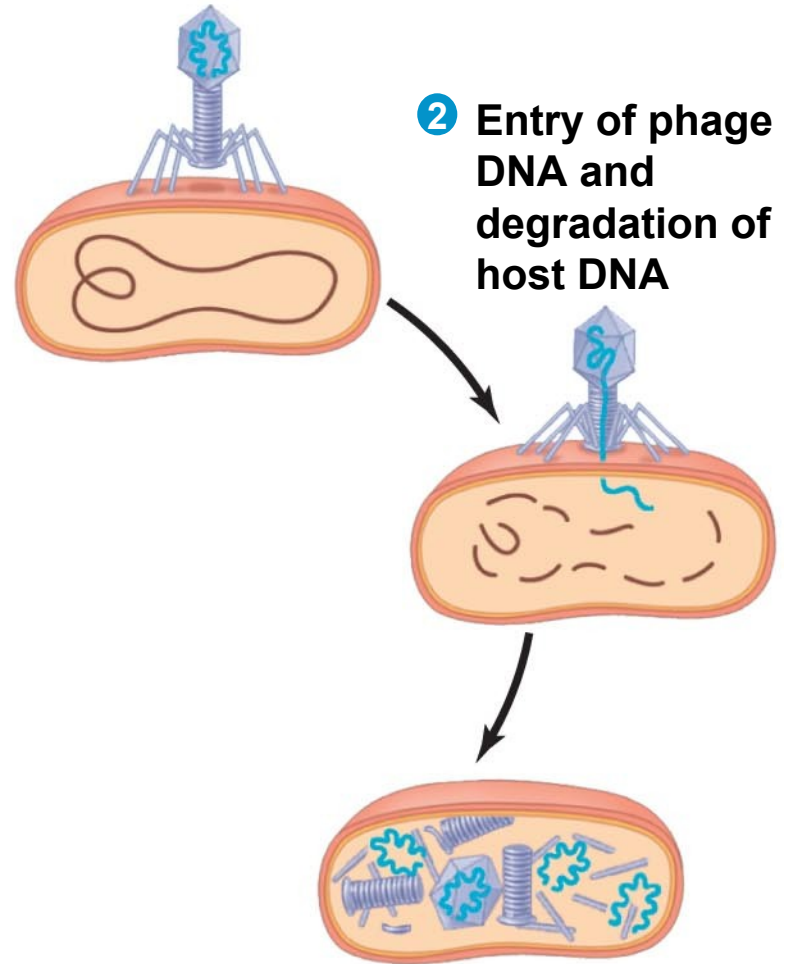
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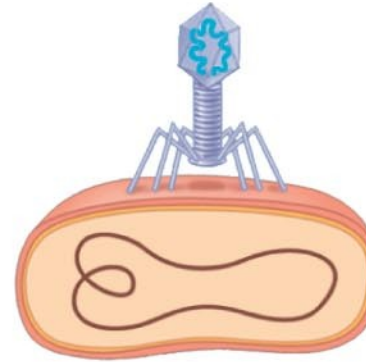
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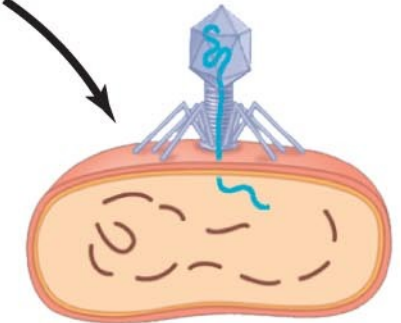
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3 Synthesis of viral genomes and proteins

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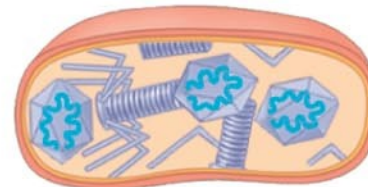
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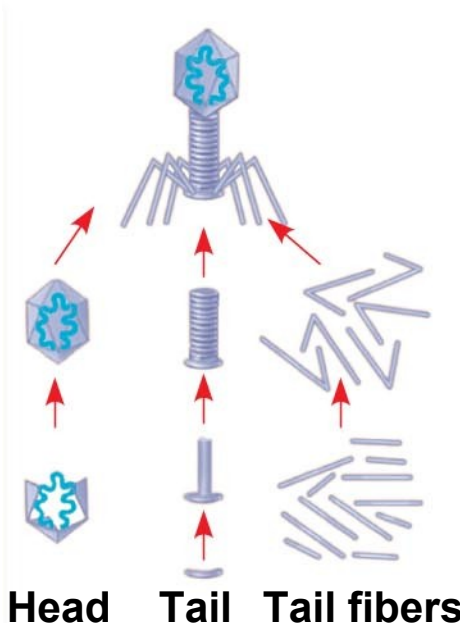
3 Synthesis of viral genomes and proteins



4 Assembly



Phage assembly



Head Tail Tail fibers

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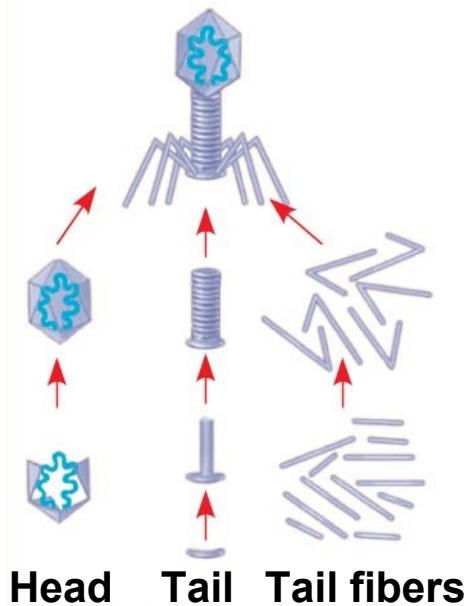
2 Entry of phage DNA and degradation of host DNA

5 Release

4 Assembly

3 Synthesis of viral genomes and proteins

Phage assembly

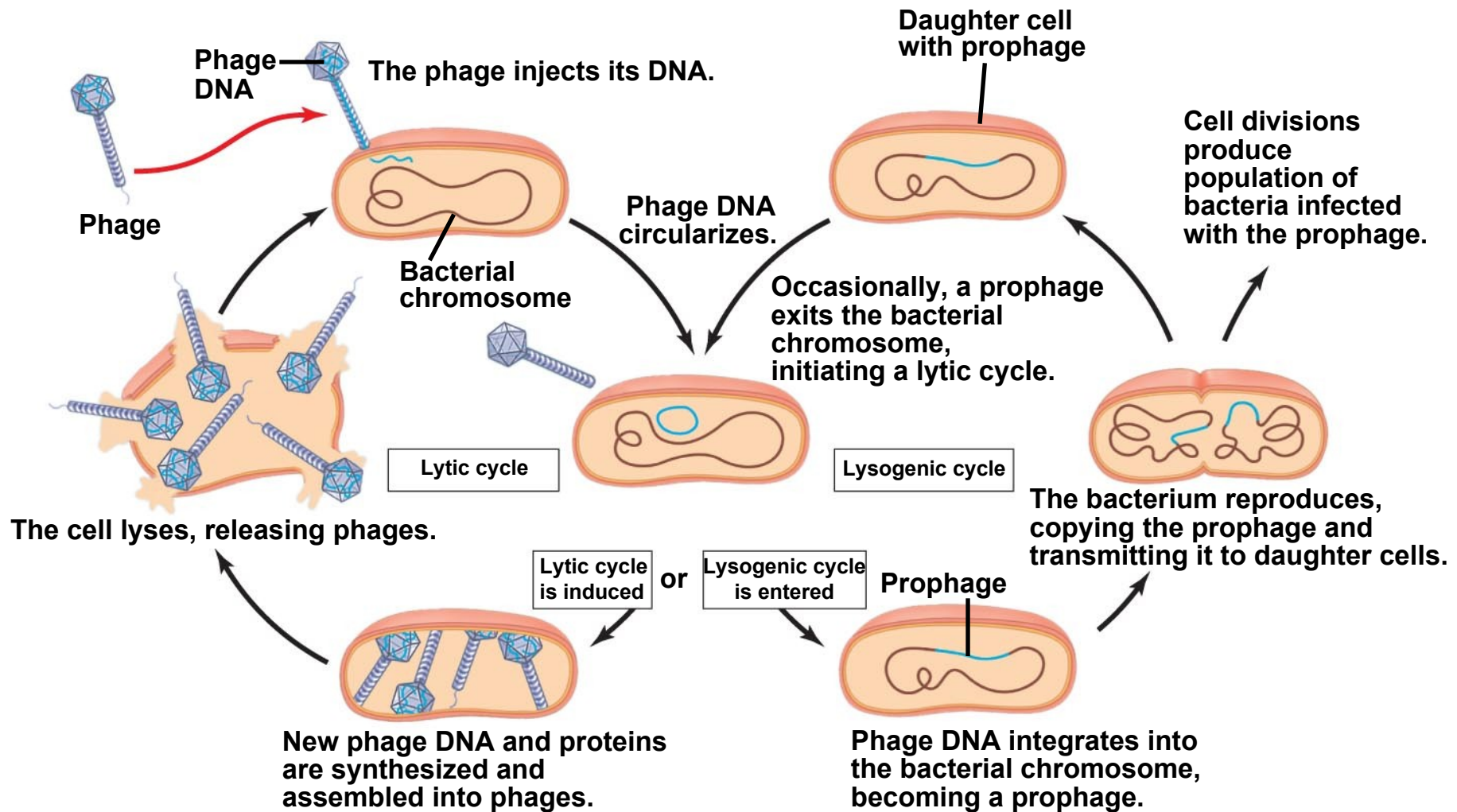


The Lysogenic Cycle

- The **lysogenic cycle** replicates the phage genome without destroying the host
- The viral DNA molecule is incorporated into the host cell's chromosome
- This integrated viral DNA is known as a **prophage**
- Every time the host divides, it copies the phage DNA and passes the copies to daughter cells

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- An environmental signal can trigger the virus genome to exit the bacterial chromosome and switch to the lytic mode
 - Phages that use both the lytic and lysogenic cycles are called **temperate phages**

Fig. 19-6



Reproductive Cycles of Animal Viruses

- There are two key variables used to classify viruses that infect animals:
 - DNA or RNA?
 - Single-stranded or double-stranded?

Table 19-1

Table 19.1 Classes of Animal Viruses		
Class/ Family	Envelope	Examples/ Disease
I. Double-stranded DNA (dsDNA)		
Adenovirus	No	Respiratory diseases; tumors
Papovavirus	No	Papillomavirus (warts, cervical cancer); polyomavirus (tumors)
Herpesvirus	Yes	Herpes simplex I and II (cold sores, genital sores); varicella zoster (shingles, chicken pox); Epstein-Barr virus (mononucleosis, Burkitt's lymphoma)
Poxvirus	Yes	Smallpox virus; cowpox virus
II. Single-stranded DNA (ssDNA)		
Parvovirus	No	B19 parvovirus (mild rash)
III. Double-stranded RNA (dsRNA)		
Reovirus	No	Rotavirus (diarrhea); Colorado tick fever virus
IV. Single-stranded RNA (ssRNA); serves as mRNA		
Picornavirus	No	Rhinovirus (common cold); poliovirus, hepatitis A virus, and other enteric (intestinal) viruses
Coronavirus	Yes	Severe acute respiratory syndrome (SARS)
Flavivirus	Yes	Yellow fever virus; West Nile virus; hepatitis C virus
Togavirus	Yes	Rubella virus; equine encephalitis viruses
V. ssRNA; template for mRNA synthesis		
Filovirus	Yes	Ebola virus (hemorrhagic fever)
Orthomyxovirus	Yes	Influenza virus
Paramyxovirus	Yes	Measles virus; mumps virus
Rhabdovirus	Yes	Rabies virus
VI. ssRNA; template for DNA synthesis		
Retrovirus	Yes	HIV, human immunodeficiency virus (AIDS); RNA tumor viruses (leukemia)

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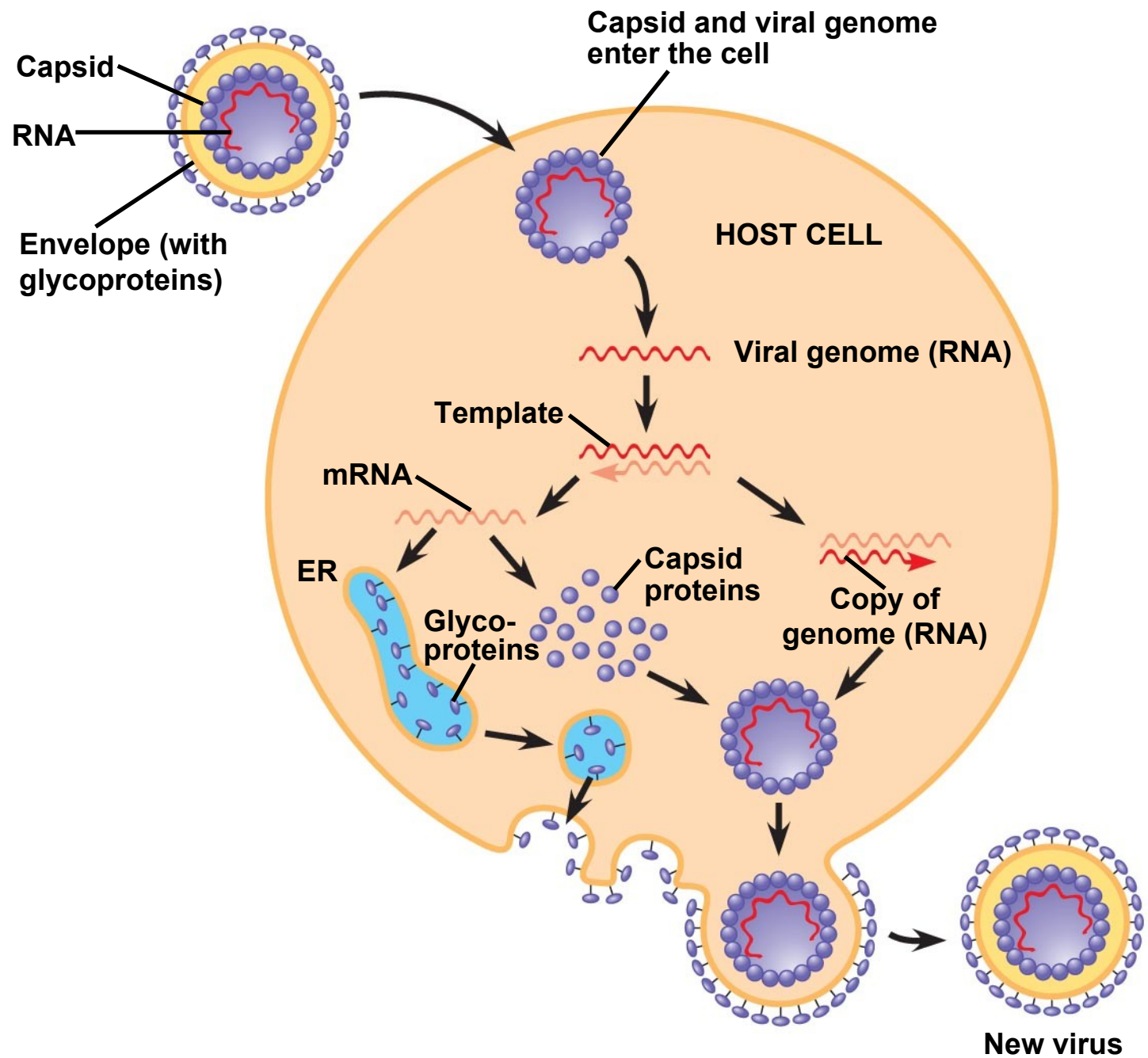
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Viral Envelopes

- Many viruses that infect animals have a membranous envelope
- Viral glycoproteins on the envelope bind to specific receptor molecules on the surface of a host cell
- Some viral envelopes are formed from the host cell's plasma membrane as the viral capsids exit

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- Other viral membranes form from the host's nuclear envelope and are then replaced by an envelope made from Golgi apparatus membrane

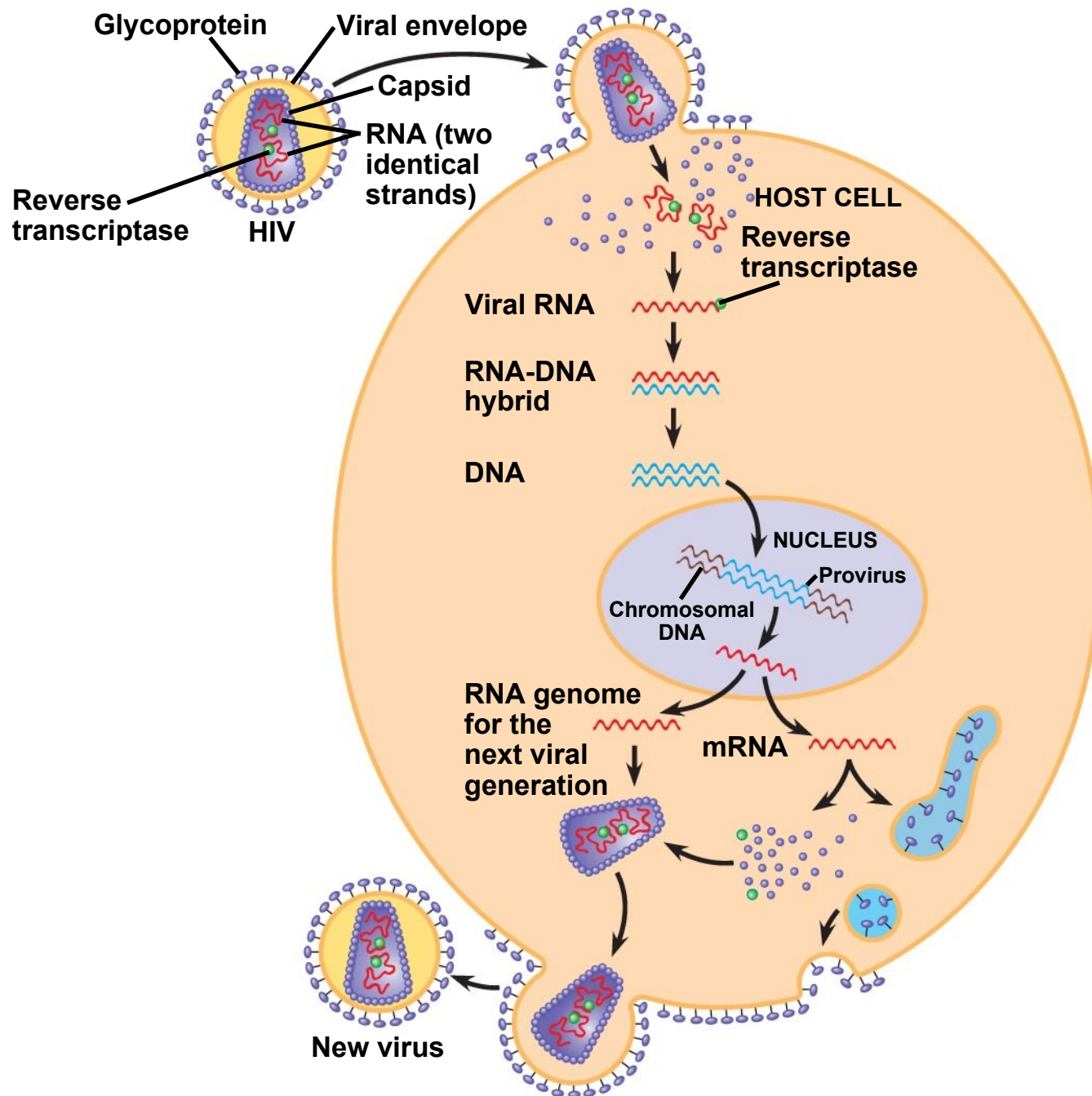
Fig. 19-7



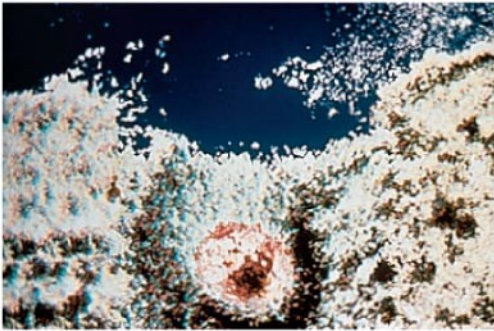
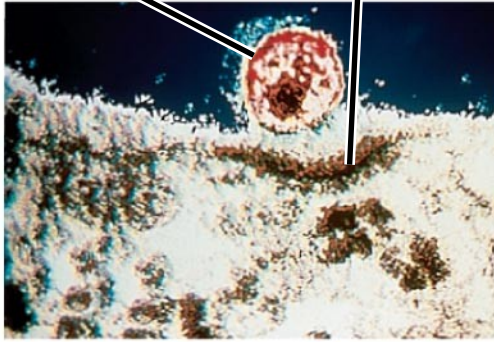
RNA as Viral Genetic Material

- The broadest variety of RNA genomes is found in viruses that infect animals
- **Retroviruses** use **reverse transcriptase** to copy their RNA genome into DNA
- **HIV (human immunodeficiency virus)** is the retrovirus that causes **AIDS (acquired immunodeficiency syndrome)**

Fig. 19-8a

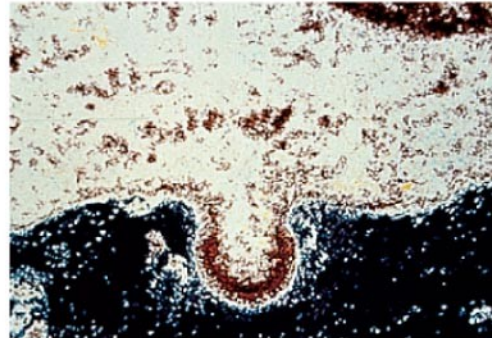
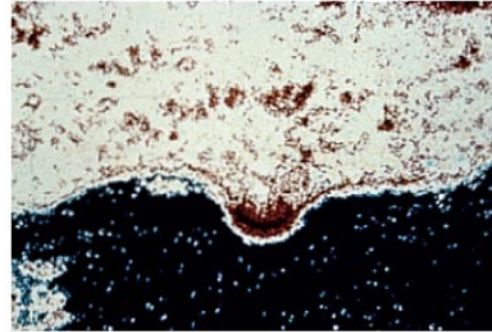


HIV
**Membrane of
white blood cell**



0.25 μm

HIV entering a cell



New HIV leaving a cell

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- The viral DNA that is integrated into the host genome is called a **provirus**
 - Unlike a prophage, a provirus remains a permanent resident of the host cell
 - The host's RNA polymerase transcribes the proviral DNA into RNA molecules
 - The RNA molecules function both as mRNA for synthesis of viral proteins and as genomes for new virus particles released from the cell

Evolution of Viruses

- Viruses do not fit our definition of living organisms
- Since viruses can reproduce only within cells, they probably evolved as bits of cellular nucleic acid
- Candidates for the source of viral genomes are plasmids, circular DNA in bacteria and yeasts, and transposons, small mobile DNA segments
- Plasmids, transposons, and viruses are all mobile genetic elements

-
- Mimivirus, a double-stranded DNA virus, is the largest virus yet discovered
 - There is controversy about whether this virus evolved before or after cells

Concept 19.3: Viruses, viroids, and prions are formidable pathogens in animals and plants

- Diseases caused by viral infections affect humans, agricultural crops, and livestock worldwide
- Smaller, less complex entities called viroids and prions also cause disease in plants and animals, respectively

Viral Diseases in Animals

- Viruses may damage or kill cells by causing the release of hydrolytic enzymes from lysosomes
- Some viruses cause infected cells to produce toxins that lead to disease symptoms
- Others have envelope proteins that are toxic

-
- **Vaccines** are harmless derivatives of pathogenic microbes that stimulate the immune system to mount defenses against the actual pathogen
 - Vaccines can prevent certain viral illnesses
 - Viral infections cannot be treated by antibiotics
 - Antiviral drugs can help to treat, though not cure, viral infections

Emerging Viruses

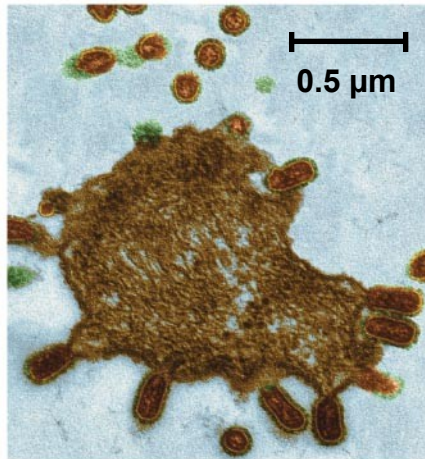
- Emerging viruses are those that appear suddenly or suddenly come to the attention of scientists
- Severe acute respiratory syndrome (SARS) recently appeared in China
- Outbreaks of “new” viral diseases in humans are usually caused by existing viruses that expand their host territory

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- Flu **epidemics** are caused by new strains of influenza virus to which people have little immunity
 - Viral diseases in a small isolated population can emerge and become global
 - New viral diseases can emerge when viruses spread from animals to humans
 - Viral strains that jump species can exchange genetic information with other viruses to which humans have no immunity

-
- These strains can cause **pandemics**, global epidemics
 - The “avian flu” is a virus that recently appeared in humans and originated in wild birds



(a) The 1918 flu pandemic



**(b) Influenza A
H5N1 virus**



(c) Vaccinating ducks

Viral Diseases in Plants

- More than 2,000 types of viral diseases of plants are known and cause spots on leaves and fruits, stunted growth, and damaged flowers or roots
- Most plant viruses have an RNA genome

Fig. 19-10

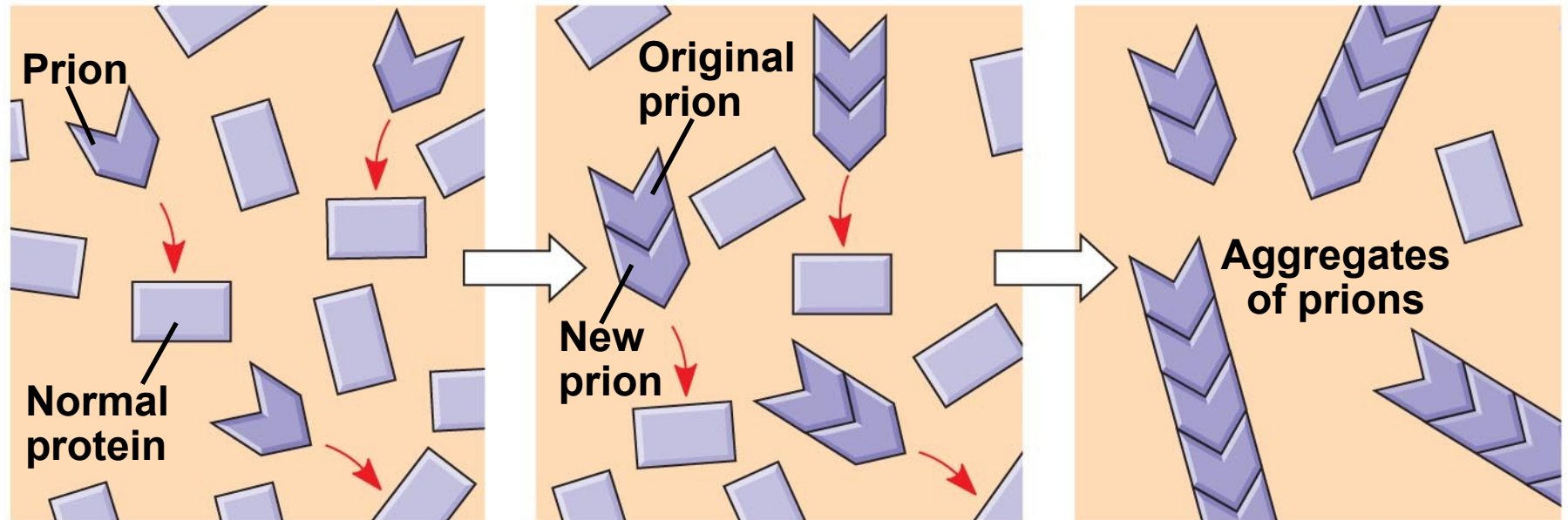


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- Plant viruses spread disease in two major modes:
 - Horizontal transmission, entering through damaged cell walls
 - Vertical transmission, inheriting the virus from a parent

Viroids and Prions: The Simplest Infectious Agents

- **Viroids** are circular RNA molecules that infect plants and disrupt their growth
- **Prions** are slow-acting, virtually indestructible infectious proteins that cause brain diseases in mammals
- Prions propagate by converting normal proteins into the prion version
- Scrapie in sheep, mad cow disease, and Creutzfeldt-Jakob disease in humans are all caused by prions

Fig. 19-11



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You should now be able to:

1. Explain how capsids and envelopes are formed
2. Distinguish between the lytic and lysogenic reproductive cycles
3. Explain why viruses are obligate intracellular parasites
4. Describe the reproductive cycle of an HIV retrovirus
5. Describe three processes that lead to the emergence of new diseases
6. Describe viroids and prions